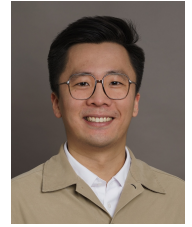


Yifeng Shen

Software Engineer

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EXPERIENCE

EDAG Engineering Group AG

Software Engineer

Mar. 2022 - Feb. 2026

Berlin, Germany

- **Autonomous driving (CityBot) - behavior & trajectory planning, Developer**
 - Developed ROS2 C++ trajectory planning module using MPC (CasADi) for real-time, obstacle-aware navigation
 - Implemented rule-based behavior planning using a finite state machine (FSM) architecture for driving decision logic
 - Designed the optimization model, including vehicle and dynamic obstacle modeling with adaptive constraints
 - Integrated multi-threaded ROS2 planning node with vehicle state, map, teleoperation, and vehicle control interfaces
 - Built MATLAB proof-of-concept with mock map data to validate and visualize planning approaches
 - Supported deployment and integration of planning software on NVIDIA Jetson AGX Xavier-based ADCU platform
- **Linux-based automotive OS (VW.OS) testing, Volkswagen Project, Component Tester**
 - Designed and maintained requirement-based test strategies across multiple product clusters
 - Developed automated test scripts using Robot Framework and integrated them into Jenkins CI pipelines (Scrum, JIRA)
 - Created and tracked defects by analyzing logs, supporting CI and daily build validation
 - Implemented shell scripts to improve testing efficiency and automation
- **Elevator HiL test automation, OTIS Project, Developer**
 - Designed and implemented an end-to-end HiL test automation pipeline for elevator control systems
 - Developed MATLAB-CANoe TCP/IP interface enabling real-time bidirectional communication
 - Implemented CAPL-based APIs in CANoe for executing control actions and retrieving system states
 - Built Python application (Tkinter GUI) for serial data parsing and integrated data into CANoe via pywin32
 - Automated Simulink test model generation from Excel and created Python scripts for execution and reporting

ContiTech, Continental AG

Intern

Apr. 2021 - Sept. 2021

Hamburg, Germany

- Designed UI/UX prototype in Figma for an RFID-based automotive connector system visualization

E.ON Energy Research Center

Master thesis

May 2020 - Jan. 2021

Aachen, Germany

- Designed and implemented IEC 61850 to XMPP communication mapping for Smart Grid applications
- Extended a gateway application with XMPP sender/receiver architecture using open-source C library (libiec61850)

NXP Cup National University Students Intelligent Car Race

Competition

June 2017 - Aug. 2017

Shanghai, China

- Developed vision-based navigation system on ARM Cortex-M4 (Kinetis K60) in C

EDUCATION

Rheinisch-Westfälische Technische Hochschule Aachen

Master of Science in Electrical Engineering, Information Technology and Computer Engineering

Oct. 2018 - Feb. 2022

Aachen, Germany

- GPA: 1.8 (best grade: 1.0)

East China University of Science and Technology

Bachelor of Engineering in Automation: Measurement and Control Technology & Instrumentation

Sept. 2014 - July 2018

Shanghai, China

- GPA: 3.75 (best grade: 4.0), Honor: National Scholarship (Awarded to top 0.2% of students in China)

SKILLS

Languages: Mandarin (Native), English (CEFR C1), German (CEFR B2)

Programming Languages: C, C++, Python, Assembly (Basic), TypeScript (Basic), PLC/Ladder Diagram (Basic)

Systems & Tools: Git, Linux, CMake, Docker, Jenkins, JIRA, Confluence, VS Code

Robotics, Embedded & Automotive: ROS2, MPC, AUTOSAR, CANoe, IEC 61850, XMPP, Real-time Systems, PLC

Testing & Validation: Robot Framework, CodeBeamer, MATLAB, Simulink, Foxglove

Web Development: Next.js, React, Tailwind CSS, Vercel, HTML5, CSS3